

A Clean Proof for Dueling Bandits

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1 Problem Setup: Contextual Dueling Bandits

At each round $t = 1, 2, \dots, T$, the learner observes a nonempty (possibly infinite) decision set

$$\mathcal{X}_t \subseteq \mathbb{R}^d, \quad \mathcal{X}_t \neq \emptyset.$$

We assume bounded features:

$$\|\mathbf{x}\|_2 \leq 1 \quad \forall \mathbf{x} \in \mathcal{X}_t, \forall t. \quad (1)$$

There exists an unknown parameter $\boldsymbol{\theta}^* \in \mathbb{R}^d$ such that the latent utility is linear:

$$f(\mathbf{x}) := \langle \boldsymbol{\theta}^*, \mathbf{x} \rangle.$$

In round t , the learner chooses a pair $(\mathbf{x}_{t,1}, \mathbf{x}_{t,2}) \in \mathcal{X}_t^2$ and observes a binary preference (reward)

$y_t \in \{0, 1\}$, where $y_t = 1$ means $\mathbf{x}_{t,1} \succ \mathbf{x}_{t,2}$, i.e., arm 1 is better than arm 2.

Define the difference feature

$$\tilde{\mathbf{x}}_t := \mathbf{x}_{t,1} - \mathbf{x}_{t,2}.$$

By (1),

$$\|\tilde{\mathbf{x}}_t\|_2 = \|\mathbf{x}_{t,1} - \mathbf{x}_{t,2}\|_2 \leq \|\mathbf{x}_{t,1}\|_2 + \|\mathbf{x}_{t,2}\|_2 \leq 2. \quad (2)$$

We assume that the observed reward y_t follows a Bernoulli distribution with the logistic/BTL model:

$$\mathbb{P}(y_t = 1 \mid \mathcal{F}_{t-1}) = \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_t \rangle), \quad \mu(z) := \frac{1}{1 + e^{-z}}, \quad (3)$$

where $\{\mathcal{F}_t\}$ is the natural filtration.

Regret. Let $\mathbf{x}_t^* \in \arg \max_{\mathbf{x} \in \mathcal{X}_t} f(\mathbf{x})$ be the optimal arm at time t . We define the (standard) instantaneous and cumulative regret for dueling bandits:

$$r_t := 2f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,1}) - f(\mathbf{x}_{t,2}), \quad R_T := \sum_{t=1}^T r_t.$$

2 Assumptions

Assumption 1 (Bounded parameter). $\|\boldsymbol{\theta}^*\|_2 \leq S$ for some known $S > 0$.

Assumption 2 (Local curvature lower bound). *There exists $\kappa_\mu > 0$ such that for all values z that can appear,*

$$\mu'(z) \geq \kappa_\mu.$$

A sufficient condition is (1) and Assumption 1, which imply $|\langle \boldsymbol{\theta}^*, \mathbf{x} - \mathbf{x}' \rangle| \leq 2S$ for any $\mathbf{x}, \mathbf{x}' \in \mathcal{X}_t$; then one may take $\kappa_\mu := \min_{|z| \leq 2S} \mu'(z) > 0$.

3 Algorithm: Ridge-MLE Dueling UCB

Algorithm 1: RIDGEMLE-DUELINGUCB

Input: Regularization $\lambda > 0$; confidence $\delta \in (0, 1)$; horizon T ;
curvature lower bound $\kappa_\mu > 0$; parameter radius $S > 0$.

- 1 Initialize $\mathbf{V}_0 \leftarrow \frac{\lambda}{\kappa_\mu} \mathbf{I}_d$
 - 2 Compute exploration radius

$$\beta_T \leftarrow \sqrt{\lambda \kappa_\mu} S + \sqrt{2 \log(1/\delta) + d \log\left(1 + \frac{4\kappa_\mu T}{\lambda d}\right)}, \quad b \leftarrow \beta_T / \kappa_\mu.$$
 - 3 **for** $t \leftarrow 1$ **to** T **do**
 - 4 Observe decision set $\mathcal{X}_t \subseteq \mathbb{R}^d$
 // (1) Compute ridge logistic MLE
 - 5 $\hat{\boldsymbol{\theta}}_t \in \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^d} \left\{ \sum_{s=1}^{t-1} \ell(y_s, \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle) + \frac{\lambda}{2} \|\boldsymbol{\theta}\|_2^2 \right\}.$
 // (2) Select two arms:
 - 6 First arm: $\mathbf{x}_{t,1} \in \arg \max_{\mathbf{x} \in \mathcal{X}_t} \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x} \rangle.$
 - 7 Second arm: $\mathbf{x}_{t,2} \in \arg \max_{\mathbf{x} \in \mathcal{X}_t} \left\{ \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x} - \mathbf{x}_{t,1} \rangle + b \|\mathbf{x} - \mathbf{x}_{t,1}\|_{\mathbf{V}_{t-1}^{-1}} \right\}.$
 // (3) Observe preference (reward)
 - 8 Observe $y_t \in \{0, 1\}.$
 // (4) Update
 - 9 $\mathbf{V}_t \leftarrow \mathbf{V}_{t-1} + \tilde{\mathbf{x}}_t \tilde{\mathbf{x}}_t^\top.$
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3.1 Derivation of the Logistic (Ridge) MLE Objective

We derive the log-likelihood and the (regularized) MLE objective.

Parametric utility. For any parameter $\boldsymbol{\theta} \in \mathbb{R}^d$, define the utility function

$$f_{\boldsymbol{\theta}}(\mathbf{x}) := \langle \boldsymbol{\theta}, \mathbf{x} \rangle.$$

The true utility function is $f(\mathbf{x}) := f_{\boldsymbol{\theta}^*}(\mathbf{x})$.

Step 1: Conditional model for a single duel. Fix a round s . The learner chooses a pair $(\mathbf{x}_{s,1}, \mathbf{x}_{s,2}) \in \mathcal{X}_s^2$ and observes $y_s \in \{0, 1\}$, where $y_s = 1$ indicates $\mathbf{x}_{s,1} \succ \mathbf{x}_{s,2}$. Under the logistic/BTL model, for any candidate parameter $\boldsymbol{\theta}$,

$$\begin{aligned}\mathbb{P}(y_s = 1 \mid \mathcal{F}_{s-1}; \boldsymbol{\theta}) &= \mu\left(f_{\boldsymbol{\theta}}(\mathbf{x}_{s,1}) - f_{\boldsymbol{\theta}}(\mathbf{x}_{s,2})\right), \\ \mathbb{P}(y_s = 0 \mid \mathcal{F}_{s-1}; \boldsymbol{\theta}) &= 1 - \mu\left(f_{\boldsymbol{\theta}}(\mathbf{x}_{s,1}) - f_{\boldsymbol{\theta}}(\mathbf{x}_{s,2})\right).\end{aligned}\tag{4}$$

Step 2: Bernoulli likelihood of one observation. Define the utility gap at time s :

$$\Delta_s(\boldsymbol{\theta}) := f_{\boldsymbol{\theta}}(\mathbf{x}_{s,1}) - f_{\boldsymbol{\theta}}(\mathbf{x}_{s,2}) = \langle \boldsymbol{\theta}, \mathbf{x}_{s,1} - \mathbf{x}_{s,2} \rangle = \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle.$$

Then the likelihood of $y_s \in \{0, 1\}$ can be written in the standard Bernoulli form, which simply combines the two equations in (4):

$$p(y_s \mid \mathcal{F}_{s-1}; \boldsymbol{\theta}) = \mu(\Delta_s(\boldsymbol{\theta}))^{y_s} (1 - \mu(\Delta_s(\boldsymbol{\theta})))^{1-y_s}.\tag{5}$$

Step 3: Conditional likelihood of the full sequence. Since the chosen pair $(\mathbf{x}_{s,1}, \mathbf{x}_{s,2})$ (hence $\tilde{\mathbf{x}}_s$) is $\mathcal{F}_{s-1}$ -measurable, the chain rule yields the factorization

$$p(y_1, \dots, y_{t-1} \mid \mathcal{F}_0; \boldsymbol{\theta}) = \prod_{s=1}^{t-1} p(y_s \mid \mathcal{F}_{s-1}; \boldsymbol{\theta}),\tag{6}$$

with each factor given by (5). The objective of MLE is to find a $\boldsymbol{\theta}$ such that (6) is maximized.

Step 4: Log-likelihood becomes a sum. Taking logs of (6) gives

$$\begin{aligned}\log p(y_1, \dots, y_{t-1} \mid \mathcal{F}_0; \boldsymbol{\theta}) &= \sum_{s=1}^{t-1} \log p(y_s \mid \mathcal{F}_{s-1}; \boldsymbol{\theta}) \\ &= \sum_{s=1}^{t-1} \left(y_s \log \mu(\Delta_s(\boldsymbol{\theta})) + (1 - y_s) \log (1 - \mu(\Delta_s(\boldsymbol{\theta}))) \right).\end{aligned}$$

Step 5: Negative log-likelihood and the logistic loss. Maximizing (6) is equivalent to minimizing the negative log-likelihood

$$-\log p(y_1, \dots, y_{t-1} \mid \mathcal{F}_0; \boldsymbol{\theta}) = \sum_{s=1}^{t-1} \ell(y_s, \Delta_s(\boldsymbol{\theta})),$$

where we define the per-round logistic loss as

$$\ell(y, z) := -y \log \mu(z) - (1 - y) \log(1 - \mu(z)).$$

Using $\Delta_s(\boldsymbol{\theta}) = \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle$, this becomes

$$\sum_{s=1}^{t-1} \ell(y_s, \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle).$$

Step 6: Ridge regularization and the ridge-MLE objective. Adding an ℓ_2 penalty yields the ridge-regularized negative log-likelihood

$$\mathcal{L}_{t-1}(\boldsymbol{\theta}) := \sum_{s=1}^{t-1} \ell(y_s, \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle) + \frac{\lambda}{2} \|\boldsymbol{\theta}\|_2^2, \quad (7)$$

and we define the ridge logistic MLE as

$$\hat{\boldsymbol{\theta}}_t \in \arg \min_{\boldsymbol{\theta} \in \mathbb{R}^d} \mathcal{L}_{t-1}(\boldsymbol{\theta}). \quad (8)$$

3.2 Derivation of the First-Order Optimality Condition

We now derive the first-order optimality condition for (8) (i.e., $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{t-1}(\boldsymbol{\theta}) = \mathbf{0}$), which will be used in the regret analysis.

Step 1: Compute $\frac{\partial}{\partial z} \ell(y, z)$. Differentiate $\ell(y, z)$:

$$\begin{aligned} \frac{\partial}{\partial z} \ell(y, z) &= -y \cdot \frac{1}{\mu(z)} \mu'(z) - (1-y) \cdot \frac{1}{1-\mu(z)} \cdot (-\mu'(z)) \\ &= \mu'(z) \left(\frac{1-y}{1-\mu(z)} - \frac{y}{\mu(z)} \right) \\ &= \mu'(z) \left(\frac{(1-y)\mu(z) - y(1-\mu(z))}{\mu(z)(1-\mu(z))} \right) \\ &= \mu'(z) \frac{\mu(z) - y}{\mu(z)(1-\mu(z))}. \end{aligned}$$

Since we assume a logistic model $\mu(z) := \frac{1}{1+e^{-z}}$, it is easy to verify that $\mu'(z) = \mu(z)(1-\mu(z))$. Using this gives the key identity:

$$\frac{\partial}{\partial z} \ell(y, z) = \mu(z) - y. \quad (9)$$

Step 2: Compute $\nabla_{\boldsymbol{\theta}} \ell(y_s, \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle)$. Since $\nabla_{\boldsymbol{\theta}} \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle = \tilde{\mathbf{x}}_s$, by the chain rule and (9),

$$\nabla_{\boldsymbol{\theta}} \ell(y_s, \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle) = \frac{\partial}{\partial z} \ell(y_s, z) \Big|_{z=\langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle} \cdot \nabla_{\boldsymbol{\theta}} \langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle = (\mu(\langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle) - y_s) \tilde{\mathbf{x}}_s.$$

Step 3: Differentiate $\mathcal{L}_{t-1}(\boldsymbol{\theta})$. Differentiate (7):

$$\nabla_{\boldsymbol{\theta}} \mathcal{L}_{t-1}(\boldsymbol{\theta}) = \sum_{s=1}^{t-1} (\mu(\langle \boldsymbol{\theta}, \tilde{\mathbf{x}}_s \rangle) - y_s) \tilde{\mathbf{x}}_s + \lambda \boldsymbol{\theta}.$$

Step 4: Stationarity at a minimizer. The function $\mathcal{L}_{t-1}(\boldsymbol{\theta})$ is convex in $\boldsymbol{\theta}$, so any minimizer $\hat{\boldsymbol{\theta}}_t$ satisfies $\nabla_{\boldsymbol{\theta}} \mathcal{L}_{t-1}(\hat{\boldsymbol{\theta}}_t) = \mathbf{0}$, yielding the first-order optimality condition:

$$\sum_{s=1}^{t-1} (\mu(\langle \hat{\boldsymbol{\theta}}_t, \tilde{\mathbf{x}}_s \rangle) - y_s) \tilde{\mathbf{x}}_s + \lambda \hat{\boldsymbol{\theta}}_t = \mathbf{0}. \quad (10)$$

4 Confidence Bound

Define the “noise”:

$$\varepsilon_s := y_s - \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle) = \begin{cases} 1 - \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle) & \text{w.p. } \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle) \\ -\mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle) & \text{w.p. } 1 - \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle) \end{cases}. \quad (11)$$

Therefore $\mathbb{E}[\varepsilon_s \mid \mathcal{F}_{s-1}] = 0$ and $|\varepsilon_s| \leq 1$, so ε_s is 1-subgaussian.

Lemma 1 (Ridge-MLE confidence ellipsoid). *Let*

$$\beta_T := \sqrt{\lambda \kappa_\mu} S + \sqrt{2 \log(1/\delta) + d \log\left(1 + \frac{4\kappa_\mu T}{\lambda d}\right)}. \quad (12)$$

With probability at least $1 - \delta$, the following holds simultaneously for all $t \leq T$:

$$\|\hat{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}} \leq \frac{\beta_T}{\kappa_\mu}. \quad (13)$$

Proof. Fix $t \leq T$ and let $\boldsymbol{\Delta}_t := \hat{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*$.

Step 1: Rewrite the stationarity condition around $\boldsymbol{\theta}^*$. Start from the first-order optimality condition (10). Add and subtract $\mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle)$ and use (11):

$$\begin{aligned} \mathbf{0} &= \sum_{s < t} (\mu(\langle \hat{\boldsymbol{\theta}}_t, \tilde{\mathbf{x}}_s \rangle) - y_s) \tilde{\mathbf{x}}_s + \lambda \hat{\boldsymbol{\theta}}_t \\ &= \sum_{s < t} (\mu(\langle \hat{\boldsymbol{\theta}}_t, \tilde{\mathbf{x}}_s \rangle) - \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle)) \tilde{\mathbf{x}}_s - \sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s + \lambda \hat{\boldsymbol{\theta}}_t. \end{aligned}$$

Rewrite $\lambda \hat{\boldsymbol{\theta}}_t = \lambda(\hat{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*) + \lambda \boldsymbol{\theta}^* = \lambda \boldsymbol{\Delta}_t + \lambda \boldsymbol{\theta}^*$ and rearrange:

$$\sum_{s < t} (\mu(\langle \hat{\boldsymbol{\theta}}_t, \tilde{\mathbf{x}}_s \rangle) - \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle)) \tilde{\mathbf{x}}_s + \lambda \boldsymbol{\Delta}_t = \sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s - \lambda \boldsymbol{\theta}^*. \quad (14)$$

For each s , by the mean value theorem, there exists some $\xi_{s,t}$ between $\langle \hat{\boldsymbol{\theta}}_t, \tilde{\mathbf{x}}_s \rangle$ and $\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle$, such that

$$\mu(\langle \hat{\boldsymbol{\theta}}_t, \tilde{\mathbf{x}}_s \rangle) - \mu(\langle \boldsymbol{\theta}^*, \tilde{\mathbf{x}}_s \rangle) = \mu'(\xi_{s,t}) \langle \boldsymbol{\Delta}_t, \tilde{\mathbf{x}}_s \rangle.$$

Plugging in (14) and multiplying $\boldsymbol{\Delta}_t^\top$ on both sides:

$$\boldsymbol{\Delta}_t^\top \left(\sum_{s < t} \mu'(\xi_{s,t}) \tilde{\mathbf{x}}_s \tilde{\mathbf{x}}_s^\top + \lambda \mathbf{I}_d \right) \boldsymbol{\Delta}_t = \boldsymbol{\Delta}_t^\top \left(\sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s - \lambda \boldsymbol{\theta}^* \right).$$

Step 2: Lower bound the left-hand side using curvature. By Assumption 2, $\mu'(\xi_{s,t}) \geq \kappa_\mu$, and by the definition of $\mathbf{V}_t$:

$$\boldsymbol{\Delta}_t^\top \left(\sum_{s < t} \mu'(\xi_{s,t}) \tilde{\mathbf{x}}_s \tilde{\mathbf{x}}_s^\top + \lambda \mathbf{I}_d \right) \boldsymbol{\Delta}_t \geq \kappa_\mu \boldsymbol{\Delta}_t^\top \left(\sum_{s < t} \tilde{\mathbf{x}}_s \tilde{\mathbf{x}}_s^\top + \frac{\lambda}{\kappa_\mu} \mathbf{I}_d \right) \boldsymbol{\Delta}_t = \kappa_\mu \|\boldsymbol{\Delta}_t\|_{\mathbf{V}_{t-1}}^2.$$

Step 3: Upper bound the right-hand side by Cauchy-Schwarz. By Cauchy-Schwarz in the $\mathbf{V}_{t-1}$ -norm,

$$\begin{aligned}\Delta_t^\top \left(\sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s - \lambda \boldsymbol{\theta}^* \right) &\leq \|\Delta_t\|_{\mathbf{V}_{t-1}} \cdot \left\| \sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s - \lambda \boldsymbol{\theta}^* \right\|_{\mathbf{V}_{t-1}^{-1}} \\ &\leq \|\Delta_t\|_{\mathbf{V}_{t-1}} \left(\left\| \sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s \right\|_{\mathbf{V}_{t-1}^{-1}} + \|\lambda \boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}^{-1}} \right).\end{aligned}$$

Combining Steps 2 and 3: When $\|\Delta_t\|_{\mathbf{V}_{t-1}} \neq 0$, cancel $\|\Delta_t\|_{\mathbf{V}_{t-1}}$ and obtain

$$\|\Delta_t\|_{\mathbf{V}_{t-1}} \leq \frac{1}{\kappa_\mu} \left(\left\| \sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s \right\|_{\mathbf{V}_{t-1}^{-1}} + \lambda \|\boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}^{-1}} \right). \quad (15)$$

If $\|\Delta_t\|_{\mathbf{V}_{t-1}} = 0$, then the claim is trivial.

Step 4: Bound the ridge term $\lambda \|\boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}^{-1}}$. Since $\mathbf{V}_{t-1} \succeq \mathbf{V}_0 = \frac{\lambda}{\kappa_\mu} \mathbf{I}_d$, we have $\mathbf{V}_{t-1}^{-1} \preceq \frac{\kappa_\mu}{\lambda} \mathbf{I}_d$, hence

$$\|\boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}^{-1}} \leq \sqrt{\frac{\kappa_\mu}{\lambda}} \|\boldsymbol{\theta}^*\|_2 \leq \sqrt{\frac{\kappa_\mu}{\lambda}} S,$$

so

$$\lambda \|\boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}^{-1}} \leq \sqrt{\lambda \kappa_\mu} S.$$

Step 5: Self-normalized bound for the noise term. Using the standard self-normalized inequality for bounded martingale differences (Theorem 1 in Abbasi-Yadkori et al. [2011]), with probability at least $1 - \delta$, simultaneously for all $t \leq T$,

$$\left\| \sum_{s < t} \varepsilon_s \tilde{\mathbf{x}}_s \right\|_{\mathbf{V}_{t-1}^{-1}} \leq \sqrt{2 \log(1/\delta) + \log \frac{\det(\mathbf{V}_{t-1})}{\det(\mathbf{V}_0)}}.$$

Next, we bound the log-determinant ratio. Since $\|\tilde{\mathbf{x}}_s\|_2 \leq 2$ by (2), by the standard determinant-trace inequality (Lemma 10 in Abbasi-Yadkori et al. [2011]) and the fact that $\det(\mathbf{V}_0) = (\frac{\lambda}{\kappa_\mu})^d$, we have

$$\log \frac{\det(\mathbf{V}_{t-1})}{\det(\mathbf{V}_0)} \leq d \log \left(1 + \frac{4\kappa_\mu T}{\lambda d} \right). \quad (16)$$

Substituting the last two displays and the ridge-term bound into (15) yields (13) with β_T defined in (12). $\square$

Corollary 1 (Difference confidence bound). *On the event of Lemma 1, for every $t \leq T$ and any $\mathbf{x}, \mathbf{x}' \in \mathcal{X}_t$,*

$$|f(\mathbf{x}) - f(\mathbf{x}') - \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x} - \mathbf{x}' \rangle| \leq b \|\mathbf{x} - \mathbf{x}'\|_{\mathbf{V}_{t-1}^{-1}}, \quad b := \beta_T / \kappa_\mu. \quad (17)$$

Proof. By definition $f(\mathbf{x}) = \langle \boldsymbol{\theta}^*, \mathbf{x} \rangle$, hence

$$f(\mathbf{x}) - f(\mathbf{x}') - \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x} - \mathbf{x}' \rangle = \langle \boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}_t, \mathbf{x} - \mathbf{x}' \rangle.$$

Apply Cauchy-Schwarz, we have:

$$|\langle \boldsymbol{\theta}^* - \hat{\boldsymbol{\theta}}_t, \mathbf{x} - \mathbf{x}' \rangle| \leq \|\hat{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}} \cdot \|\mathbf{x} - \mathbf{x}'\|_{\mathbf{V}_{t-1}^{-1}}.$$

Using Lemma 1, $\|\hat{\boldsymbol{\theta}}_t - \boldsymbol{\theta}^*\|_{\mathbf{V}_{t-1}} \leq b$, yielding (17). $\square$

5 Regret Analysis

In this section we work on the high-probability event of Corollary 1. Define

$$U_t(\mathbf{x} \mid \mathbf{x}_{t,1}) := \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x} - \mathbf{x}_{t,1} \rangle + b \|\mathbf{x} - \mathbf{x}_{t,1}\|_{\mathbf{V}_{t-1}^{-1}}. \quad (18)$$

We use the key fact that $\mathbf{x}_{t,2}$ maximizes $U_t(\cdot \mid \mathbf{x}_{t,1})$ and that $\mathbf{x}_{t,1}$ maximizes $\langle \hat{\boldsymbol{\theta}}_t, \cdot \rangle$ over $\mathcal{X}_t$.

Lemma 2 (Optimal to first arm). *For every $t \leq T$,*

$$f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,1}) \leq b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}.$$

Proof. By Corollary 1 with $(\mathbf{x}, \mathbf{x}') = (\mathbf{x}_t^*, \mathbf{x}_{t,1})$,

$$f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,1}) \leq U_t(\mathbf{x}_t^* \mid \mathbf{x}_{t,1}).$$

Since $\mathbf{x}_{t,2} \in \arg \max_{\mathbf{x} \in \mathcal{X}_t} U_t(\mathbf{x} \mid \mathbf{x}_{t,1})$,

$$U_t(\mathbf{x}_t^* \mid \mathbf{x}_{t,1}) \leq U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1}).$$

Finally, because $\mathbf{x}_{t,1} \in \arg \max_{\mathbf{x} \in \mathcal{X}_t} \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x} \rangle$, we have $\langle \hat{\boldsymbol{\theta}}_t, \mathbf{x}_{t,2} - \mathbf{x}_{t,1} \rangle \leq 0$, hence

$$U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1}) = \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x}_{t,2} - \mathbf{x}_{t,1} \rangle + b \|\mathbf{x}_{t,2} - \mathbf{x}_{t,1}\|_{\mathbf{V}_{t-1}^{-1}} \leq b \|\mathbf{x}_{t,2} - \mathbf{x}_{t,1}\|_{\mathbf{V}_{t-1}^{-1}} = b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}.$$

$\square$

Lemma 3 (Optimal to second arm). *For every $t \leq T$,*

$$f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,2}) \leq 2b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}.$$

Proof. We use the identity

$$f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,2}) = (f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,1})) + (f(\mathbf{x}_{t,1}) - f(\mathbf{x}_{t,2})).$$

We bound the two terms in a way that makes $U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1})$ cancel.

Step 1: Upper bound $f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,1})$ by $U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1})$. By Corollary 1 with $(\mathbf{x}, \mathbf{x}') = (\mathbf{x}_t^*, \mathbf{x}_{t,1})$ and the fact that $\mathbf{x}_{t,2} \in \arg \max_{\mathbf{x} \in \mathcal{X}_t} U_t(\mathbf{x} \mid \mathbf{x}_{t,1})$,

$$f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,1}) \leq U_t(\mathbf{x}_t^* \mid \mathbf{x}_{t,1}) \leq U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1}).$$

Step 2: Upper bound $f(\mathbf{x}_{t,1}) - f(\mathbf{x}_{t,2})$ with an explicit $-U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1})$ term. Apply Corollary 1 with $(\mathbf{x}, \mathbf{x}') = (\mathbf{x}_{t,1}, \mathbf{x}_{t,2})$:

$$f(\mathbf{x}_{t,1}) - f(\mathbf{x}_{t,2}) \leq \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x}_{t,1} - \mathbf{x}_{t,2} \rangle + b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}.$$

Rewrite $\langle \hat{\boldsymbol{\theta}}_t, \mathbf{x}_{t,1} - \mathbf{x}_{t,2} \rangle = -\langle \hat{\boldsymbol{\theta}}_t, \mathbf{x}_{t,2} - \mathbf{x}_{t,1} \rangle$ and add/subtract $b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}$:

$$\begin{aligned} \langle \hat{\boldsymbol{\theta}}_t, \mathbf{x}_{t,1} - \mathbf{x}_{t,2} \rangle + b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} &= 2b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} - \left(\langle \hat{\boldsymbol{\theta}}_t, \mathbf{x}_{t,2} - \mathbf{x}_{t,1} \rangle + b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} \right) \\ &= 2b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} - U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1}). \end{aligned}$$

Therefore,

$$f(\mathbf{x}_{t,1}) - f(\mathbf{x}_{t,2}) \leq 2b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} - U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1}).$$

Step 3: Add Step 1 and Step 2 and cancel $U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1})$. Combining,

$$f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,2}) \leq U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1}) + \left(2b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} - U_t(\mathbf{x}_{t,2} \mid \mathbf{x}_{t,1}) \right) = 2b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}.$$

□

Lemma 4 (Instantaneous regret). *For every $t \leq T$,*

$$r_t \leq 3b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}.$$

Proof. By definition,

$$r_t = (f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,1})) + (f(\mathbf{x}_t^*) - f(\mathbf{x}_{t,2})).$$

Apply Lemma 2 and Lemma 3 and add:

$$r_t \leq b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} + 2b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} = 3b \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}.$$

□

Lemma 5 (Elliptical potential bound).

$$\sum_{t=1}^T \min \left\{ 1, \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}^2 \right\} \leq 2d \log \left(1 + \frac{4\kappa_\mu T}{\lambda d} \right).$$

Proof. We first relate $\det(\mathbf{V}_t)$ to $\det(\mathbf{V}_{t-1})$ using the fact that for any invertible matrix $\mathbf{A} \in \mathbb{R}^{d \times d}$ and any vectors $\mathbf{u}, \mathbf{v} \in \mathbb{R}^d$,

$$\det(\mathbf{A} + \mathbf{u}\mathbf{v}^\top) = \det(\mathbf{A})(1 + \mathbf{v}^\top \mathbf{A}^{-1} \mathbf{u}).$$

Let $\mathbf{A} = \mathbf{V}_{t-1}$ and $\mathbf{u} = \mathbf{v} = \tilde{\mathbf{x}}_t$. Since $\mathbf{V}_{t-1}$ is invertible,

$$\det(\mathbf{V}_t) = \det(\mathbf{V}_{t-1}) \left(1 + \tilde{\mathbf{x}}_t^\top \mathbf{V}_{t-1}^{-1} \tilde{\mathbf{x}}_t\right) = \det(\mathbf{V}_{t-1}) \left(1 + \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}^2\right). \quad (19)$$

Iterating (19) from $t = 1$ to T yields

$$\det(\mathbf{V}_T) = \det(\mathbf{V}_0) \prod_{t=1}^T \left(1 + \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}^2\right). \quad (20)$$

Divide by $\det(\mathbf{V}_0)$ and take logarithms:

$$\log \frac{\det(\mathbf{V}_T)}{\det(\mathbf{V}_0)} = \sum_{t=1}^T \log \left(1 + \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}^2\right). \quad (21)$$

Since $\min\{1, x\} \leq 2 \log(1 + x)$ for $x \geq 0$, by (21),

$$\begin{aligned} \sum_{t=1}^T \min\left\{1, \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}^2\right\} &\leq 2 \sum_{t=1}^T \log \left(1 + \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}^2\right) \\ &= 2 \log \frac{\det(\mathbf{V}_T)}{\det(\mathbf{V}_0)} \leq 2d \log \left(1 + \frac{4\kappa_\mu T}{\lambda d}\right). \end{aligned}$$

where the last inequality is due to (16). $\square$

6 Main Result: High-Probability Regret Bound

Theorem 1 (High-probability regret bound). *Under Assumptions 1 and 2, with probability at least $1 - \delta$,*

$$R_T \leq 3b \sum_{t=1}^T \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}} \leq 3b \sqrt{2Td \log \left(1 + \frac{4\kappa_\mu T}{\lambda d}\right)}.$$

Equivalently,

$$R_T = \tilde{\mathcal{O}}\left(\frac{d}{\kappa_\mu} \sqrt{T}\right),$$

where $\tilde{\mathcal{O}}(\cdot)$ hides logarithmic factors in T , d , and $1/\delta$.

Proof. We work on the event of Corollary 1, which holds with probability at

least $1 - \delta$ by Lemma 1. Summing Lemma 4 over $t = 1, \dots, T$ gives

$$\begin{aligned} R_T &= \sum_{t=1}^T r_t \\ &\leq \sum_{t=1}^T \min\{1, r_t\} \end{aligned} \tag{22}$$

$$\leq 3b \sum_{t=1}^T \min\{1, \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}\} \tag{23}$$

$$\leq 3b \sqrt{T \sum_{t=1}^T \min\left\{1, \|\tilde{\mathbf{x}}_t\|_{\mathbf{V}_{t-1}^{-1}}^2\right\}} \tag{24}$$

$$\leq 3b \sqrt{2Td \log\left(1 + \frac{4\kappa_\mu T}{\lambda d}\right)}, \tag{25}$$

where (22) uses the fact that $r_t \leq 1$ (**This is not fully true... According to the definition, it should be $r_t \leq 4S$, but Abbasi-Yadkori et al. [2011] also uses a similar assumption.**), (23) is due to Lemma 4, (24) uses the Cauchy–Schwarz inequality, and (25) uses Lemma 5. $\square$

References

Yasin Abbasi-Yadkori, Dávid Pál, and Csaba Szepesvári. Improved algorithms for linear stochastic bandits. In *Proceedings of the 24th International Conference on Neural Information Processing Systems, NIPS’11*, page 2312–2320, 2011.